

Harsh Kumar

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RESEARCH INTERESTS

I study how everyday use of LLMs shapes people's learning, creativity, and wellbeing. Evaluations today largely capture what models can do, and increasingly what humans and AI accomplish together. My work asks about a harder-to-measure layer, the lasting effects of these interactions on human capacities, which I study through large-scale randomized experiments and other methods from social science. I use those findings to design better evaluations and post-training methods for AI systems.

EDUCATION

University of Toronto

Toronto, ON

Ph.D. in Computer Science (expected Fall 2026)

September 2021 – Present

Advisor: Ashton Anderson. Selected courses: Machine Learning, Computational Social Science, AI Alignment.

- Dissertation: *Human Capacities in the Age of AI: Designing and Evaluating LLM Systems for Learning, Creativity, and Wellbeing.*

Vellore Institute of Technology

Tamil Nadu, India

B.Tech. in Computer Science and Engineering (CGPA: 8.99/10)

July 2016 – July 2020

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Research Assistant

September 2021 – Present

University of Toronto (Advisor: Ashton Anderson)

Toronto, ON

- Conducted large-scale controlled and field experiments on how everyday LLM use affects learning, creativity, and wellbeing, including online studies with 1,100+ participants and randomized deployments in university classrooms [C2, C7, C9, C11].
- Developed the software infrastructure to run these experiments, including study websites and text-messaging systems used by thousands of participants.
- Personalized education and mental-health interventions with contextual bandits, reaching 10,000+ young adults through Mental Health America's text-messaging program [C12], and built LLM agents for behavior-change coaching [C4].
- Found that frontier LLMs collapse over 90% of advice contexts into a single supportive default; built the post-training repair with supervised fine-tuning of open-weight models (inverse-process distillation) and validated it with automated evals and an approximately 200-participant human evaluation [P1].

Applied Scientist Intern

May 2025 – August 2025

Microsoft, Viva Insights (with Shamsi Iqbal)

Redmond, WA

- Designed and evaluated LLM-based agents that coach organizational leaders through AI adoption, helping them interpret adoption metrics and identify risks and opportunities for sustainable deployment [P3].
- Validated the designs through leader interviews, a large controlled experiment, and in-situ studies with managers; translated findings into guidance for communicating AI-adoption metrics to decision-makers.

Research Intern

Summers 2023 & 2024

Microsoft Research, Computational Social Science group

New York City, NY

- Ran two of the first large, pre-registered randomized experiments on LLM math tutoring (N = 1,818 U.S. adults), building the GPT-4 and Phi-3 tutoring and evaluation pipelines [C6].
- Found that LLM explanations improved unassisted test performance over correct answers alone, especially when participants attempted problems on their own first, and that even LLM explanations containing arithmetic errors produced more learning than answers alone.

Software Engineer

January 2020 – August 2021

J.P. Morgan Chase & Co.

Mumbai, India

- Built and shipped a production NLU system powering internal chatbots for investment bankers, with user-feedback-driven improvement loops (training, deployment, CI/CD).

Research Analyst Intern

May 2019 – January 2020

McKinsey & Company

Gurugram, India

- Developed the vision system (visual SLAM) for a semi-autonomous UAV built to optimize warehouse operations.

* denotes equal contribution.

- [C1] Ashton Anderson, **Harsh Kumar**, Louis Tay, Karina Vold. *Position: We Need Large Language Models Optimized For Our Well-Being*. **ICML 2026**. [PDF] [poster]
- [C2] **Harsh Kumar**, Jasmine Chahal, Yinuo Zhao, Zoey Zhang, Annika Wei, Louis Tay, Ashton Anderson. *When AI Gives Advice: Evaluating AI and Human Responses to Online Advice-Seeking for Well-Being*. **CHI 2026**. [doi:10.1145/3772318.3791233]
- [C3] Jiayin Zhi, **Harsh Kumar**, Mina Lee. *Investigating the Effects of LLM Use on Critical Thinking under Time Constraints: Access Timing and Time Availability*. **CHI 2026**. [doi:10.1145/3772318.3791796]
- [C4] **Harsh Kumar**, SuHyeon Yoo, Angela Zavaleta Bernuy, Jiakai Shi, Huayin Luo, Joseph Jay Williams, Anastasia Kuzminykh, Ashton Anderson, Rachel Kornfield. *Large Language Model Agents for Improving Engagement with Behavior Change Interventions: Application to Digital Mindfulness*. **PACMHCI (CSCW), 2025**. [doi:10.1145/3757619]
- [C5] Ananya Bhattacharjee, Joseph Jay Williams, Miranda Beltzer, Jonah Meyerhoff, **Harsh Kumar**, Haochen Song, David C. Mohr, Alex Mariakakis, Rachel Kornfield. *Investigating the Role of Situational Disruptors in Engagement with Digital Mental Health Tools*. **PACMHCI (CSCW), 2025. Honorable Mention Award (top 5%)**. [doi:10.1145/3757487]
- [C6] **Harsh Kumar**, David M. Rothschild, Daniel G. Goldstein, Jake M. Hofman. *Math Education With Large Language Models: Peril or Promise?* **AIED 2025**. [link]
- [C7] **Harsh Kumar**, Jonathan Vincentius, Ewan Jordan, Ashton Anderson. *Human Creativity in the Age of LLMs: Randomized Experiments on Divergent and Convergent Thinking*. **CHI 2025. Honorable Mention Award (top 5%)**. [doi:10.1145/3706598.3714198]
- [C8] Ilya Musabirov, Mohi Reza, Haochen Song, Steven Moore, Pan Chen, **Harsh Kumar**, Tong Li, John Stamper, Norman Bier, Anna Rafferty, Thomas Price, Nina Deliu, Audrey Durand, Michael Liut, Joseph Jay Williams. *Platform-based Adaptive Experimental Research in Education: Lessons Learned from The Digital Learning Challenge*. **LAK 2025**. [doi]
- [C9] **Harsh Kumar**, Ilya Musabirov, Mohi Reza, Jiakai Shi, Joseph J. Williams, Anastasia Kuzminykh, Michael Liut. *Guiding Students in Using LLMs in Supported Learning Environments: Effects on Interaction Dynamics, Learner Performance, Confidence, and Trust*. **PACMHCI (CSCW), 2024**. [doi:10.1145/3687038]
- [C10] Jessica Y. Bo*, **Harsh Kumar***, Michael Liut, Ashton Anderson. *Disclosures & Disclaimers: Investigating the Impact of Transparency Disclosures and Reliability Disclaimers on Learner-LLM Interactions*. **HCOMP 2024**. [link]
- [C11] **Harsh Kumar**, Ruiwei Xiao, Benjamin Lawson, Ilya Musabirov, Jiakai Shi, Xinyuan Wang, Huayin Luo, Joseph J. Williams, Anna Rafferty, John Stamper, Michael Liut. *Supporting Self-Reflection at Scale with Large Language Models: Insights from Randomized Field Experiments in Classrooms*. **L@S 2024**. [doi:10.1145/3657604.3662042]
- [C12] **Harsh Kumar**, Tong Li, Jiakai Shi, Ilya Musabirov, Rachel Kornfield, Jonah Meyerhoff, Ananya Bhattacharjee, Chris Karr, David Mohr, Anna Rafferty, Sofia Villar, Nina Deliu, Joseph Jay Williams. *Using Adaptive Bandit Experiments to Increase and Investigate Engagement in Digital Mental Health*. **AAAI 2024 (IAAI)**. [link]
- [C13] Ananya Bhattacharjee, Joseph J. Williams, Jonah Meyerhoff, **Harsh Kumar**, Alex Mariakakis, Rachel Kornfield. *Investigating the Role of Context in the Delivery of Text Messages for Supporting Psychological Wellbeing*. **CHI 2023. Best Paper Award (top 1%)**. [doi]
- [C14] Rachel Kornfield, Caitlin A. Stamatis, Ananya Bhattacharjee, Bei Pang, Theresa Nguyen, Joseph J. Williams, **Harsh Kumar**, Sarah Popowski, Miranda Beltzer, Christopher J. Karr, Madhu Reddy, David C. Mohr, Jonah Meyerhoff. *A Text Messaging Intervention to Support the Mental Health of Young Adults: User Engagement and Feedback from a Field Trial of an Intervention Prototype*. **Internet Interventions, 2023**. [link]
- [C15] Jonah Meyerhoff, Miranda Beltzer, Sarah Popowski, Chris J. Karr, Theresa Nguyen, Joseph J. Williams, Charles J. Krause, **Harsh Kumar**, Ananya Bhattacharjee, David C. Mohr, Rachel Kornfield. *Small Steps over Time: A Longitudinal Usability Test of an Automated Interactive Text Messaging Intervention to Support Self-Management of Depression and Anxiety Symptoms*. **Journal of Affective Disorders, 2023**. [link]

WORKING PAPERS & UNDER REVIEW

- [P1] **Harsh Kumar**, Karina Vold, Louis Tay, Ashton Anderson. *Diagnosing and Repairing Persona Collapse in LLM Advice*. Under review, 2026. [arXiv:2607.08326]
- [P2] **Harsh Kumar**, Jace Mu, Jonathan Vincentius, Ashton Anderson. *Human Thinking under Plural LLM Assistance: Mathematical Problem Solving and Open-Ended Writing*. Under review, 2026. [arXiv:2604.02677]
- [P3] **Harsh Kumar**, Christopher M. Gideon, Nick DeFalco, Alyssa Rizzolo, Adam Coleman, Shamsi T. Iqbal. *From Metrics to Decisions: Designing LLM Support for Managerial Sensemaking About Organizational AI Adoption*. Under submission (work done during an internship at Microsoft).
- [P4] Daniel R. Harris, Karina Vold, **Harsh Kumar**, Ashton Anderson. *A Taxonomy of Value Alignment Problems: Subproblems and Solutions*. Under review, 2026.
- [P5] Ruiwei Xiao, Runlong Ye, Xinying Hou, Jionghao Wen, **Harsh Kumar**, Michael Liut, John Stamper. *Transforming GenAI Policy to Prompting Instruction: An RCT of Scalable Prompting Interventions in a CS1 Course*. Under review, 2026. [arXiv:2602.16033]
- [P6] Sofia Panasiuk, Dana Kulzhabayeva, Mohi Reza, **Harsh Kumar**, Ananya Bhattacharjee, Felix Cheung, Joseph Jay Williams. *Doing Team Science at Scale: Practical Guidelines for Leading a Crowdsourced Scientific Collaboration in Psychology*. Under review at AMPPS. [OSF preprint]
- [P7] **Harsh Kumar**, Ilya Musabirov, Jiakai Shi, Adele Lauzon, Kwan Kiu Choy, Ofek Gross, Dana Kulzhabayeva, Joseph J. Williams. *Exploring The Design of Prompts For Applying GPT-3 based Chatbots: A Mental Wellbeing Case Study on Mechanical Turk*. [arXiv:2209.11344]. Presented at CODE@MIT 2022.
- [P8] Tong Li, Jacob Nogas, Haochen Song, **Harsh Kumar**, Audrey Durand, Anna Rafferty, Nina Deliu, Sofia S. Villar, Joseph J. Williams. *Algorithms for Adaptive Experiments that Trade-off Statistical Analysis with Reward*. [arXiv:2112.08507]

WORKSHOP PAPERS, POSTERS & EXTENDED ABSTRACTS

- [W1] **Harsh Kumar**, Mohi Reza, Jeb Thomas-Mitchell, Ilya Musabirov, Lisa Zhang, Michael Liut. *Understanding Help-Seeking Behavior of Students Using LLMs vs. Web Search for Writing SQL Queries*. DataEd Workshop, 2025. [doi]
- [W2] Ruiwei Xiao, Xinying Hou, **Harsh Kumar**, Steven Moore, John Stamper, Michael Liut. *A Preliminary Analysis of Students' Help Requests with an LLM-powered Chatbot when Completing CS1 Assignments*. CSEDM Workshop @ EDM, 2024. [link]
- [W3] **Harsh Kumar**, Yiyi Wang, Jiakai Shi, Ilya Musabirov, Norman A. S. Farb, Joseph J. Williams. *Exploring the Use of Large Language Models for Improving the Awareness of Mindfulness*. Extended Abstract, CHI 2023. [doi]
- [W4] **Harsh Kumar**, Kunzhi Yu, Andrew Chung, Jiakai Shi, Joseph J. Williams. *Exploring The Potential of Chatbots to Provide Mental Well-being Support for Computer Science Students*. Poster, SIGCSE TS 2023. [doi]
- [W5] **Harsh Kumar**, Ilya Musabirov, Joseph J. Williams, Michael Liut. *QuickTA: Exploring the Design Space of Using Large Language Models to Provide Support to Students*. LAK Workshop, 2023. [link]
- [W6] **Harsh Kumar**, Dana Kulzhabayeva, Joseph J. Williams. *Using Adaptive Experiments for Personalizing Mental-Health Related Interventions to Account for Heterogeneity between Participants*. Poster, Intervention Science Preconference, SPSP 2023.
- [W7] **Harsh Kumar***, Taneea S. Agrawaal*, Kwan Kiu Choy, Jiakai Shi, Joseph J. Williams. *"Sounds like a Cheesy Radio Ad": Using User Perspectives for Enhancing Digital COVID Vaccine Communication Strategies for Public Health Agencies*. Extended Abstract, CHI 2022. [doi]
- [W8] **Harsh Kumar**, Jiakai Shi, Ilya Musabirov, Adele Lauzon, Meagan Peters, Ofek Gross, Joseph J. Williams. *Prompt Engineering to Improve Experiences with Large Language Model-Administered Support*. Parallel Talk, CODE 2022.
- [W9] Aayush Kapur, Himanshu Thakur, **Harsh Kumar**. *Safeguarding People against Social Media Frauds during the COVID-19 Oxygen Supply Crisis in India*. Poster, ML4D Workshop @ NeurIPS 2021. [OSF]

AWARDS, GRANTS & COMPETITIONS

XPRIZE Digital Learning Challenge — US\$500,000 Grand Prize

April 2023

Adaptive Experimentation Accelerator team [announcement]

- Global competition to modernize and accelerate the identification of effective learning tools; led development of the contextual-bandit personalization system (MOOClet) enabling large-scale adaptive experiments in classrooms.

DARPA AI Tools for Adult Learning — US\$250,000 (project: QuickTA)

September 2023

[announcement]

- LLM-based learning tool combining curated educational content, adaptive RL algorithms, and intelligent tutoring systems; has supported 5,000+ students at the University of Toronto. Co-led with John Stamper (CMU) and Norman Bier (Open Learning Initiative).

Florea.ai — John Templeton Foundation grant on wellbeing and AI

PIs: Louis Tay, Karina Vold, Ashton Anderson [site]

- Contributed to the funded grant proposal; lead the technical side of the project as the senior graduate student.

Best Paper Award (top 1%), CHI 2023 • **Honorable Mention** (top 5%), CHI 2025, CSCW 2025.

Research covered by **Time**, **The Economist**, **The New York Times**, **Forbes**, and **Science News**; featured in Microsoft's *New Future of Work* report.

TECHNICAL SKILLS

LLM training & evaluation: supervised fine-tuning (full / LoRA), Hugging Face Transformers & TRL, vLLM inference, multi-GPU training (SLURM clusters), LLM-as-judge and human-evaluation design, benchmark construction
Experimentation: randomized experiments / A-B testing, causal inference, adaptive experiments (contextual bandits), crowdsourced studies (Prolific, MTurk), statistical analysis in R and Python, frontend development using React
Engineering: Python, PyTorch, R, SQL, JavaScript; React, Flask/Django, Docker, CI/CD; production ML systems

TEACHING

Teaching Assistant, University of Toronto

September 2021 – Present

Department of Computer Science

- Courses: CSC428 (HCI), CSC2558 (Multidisciplinary HCI), CSC2552 (Computational Social Science), CSC318 (Design of Interactive Media), CSC343 (Databases), CSC108 (Intro Programming).
- Led weekly tutorial sections; facilitated in-class exercises and discussions; supported students on experimental design, measurement, and debugging; assisted with grading and course administration.

MENTORSHIP

Undergraduate and graduate research mentees at the University of Toronto, including: Jiakai Shi, Kwan Kiu (Koby) Choy, Jonathan Vincentius, Ewan Jordan, Jace Mu, Jasmine Chahal, Yinuo Zhao, Zoey Zhang, Mieke Zhao.

ACADEMIC SERVICE

Peer Reviewer

CHI, CSCW, UIST, DIS, IUI, Creativity & Cognition, The Web Conference, AAAI, AIES, HCOMP, L@S, AIED

Expert Reviewer, Tools Competition

January 2024 & January 2025

Multi-million-dollar EdTech funding competition; reviewed and advised qualified teams

Session Chair (Track: AI and HCI)

July 2024

10th International Conference on Computational Social Science (IC2S2), Philadelphia

Workshop Co-organizer, CHI 2023

April 2023

Integrating Individual and Social Contexts into Self-Reflection Technologies [site]

REFERENCES

- Dr. Ashton Anderson, Associate Professor, Computer Science, University of Toronto — ashton@cs.toronto.edu
- Dr. Daniel G. Goldstein, Partner Research Manager, Microsoft Research NYC — dgg@microsoft.com
- Dr. Jake M. Hofman, Senior Principal Researcher, Microsoft Research NYC — jmh@microsoft.com
- Dr. David M. Rothschild, Economist, Microsoft Research NYC — david@researchdmr.com
- Dr. Louis Tay, William C. Byham Professor, Industrial-Organizational Psychology, Purdue University — stay@purdue.edu
- Dr. Shamsi Iqbal, Senior Principal Research Manager, Microsoft — shamsi@microsoft.com
- Dr. Anastasia Kuzminykh, Assistant Professor, Faculty of Information, University of Toronto — anastasia.kuzminykh@utoronto.ca
- Dr. Michael Liut, Associate Professor (Teaching Stream), MCS, University of Toronto Mississauga — michael.liut@utoronto.ca
- Dr. Rachel Kornfield, Research Assistant Professor, Feinberg School of Medicine, Northwestern University — rachel.kornfield@northwestern.edu